Windows Update fails or gets stuck: fixes

Article KB0010045 | Owner: Desktop Support | Subcategory: OS update failure | Applies to: Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, Windows 11 22H2, 23H2, 24H2, Devices managed by Microsoft Intune

Summary

Windows Update on Acme Global-managed laptops can fail or appear stuck, showing errors such as 0x80070002, 0% progress, or continuous roll-back. Most issues are resolved by running the built-in troubleshooter, resetting update components, freeing disk space, ensuring VPN connectivity, and, if needed, installing the latest cumulative update manually from the Microsoft Update Catalog. This article provides end-user self-help steps and a detailed service-desk procedure to diagnose and fix the problem.

Symptoms

- Windows Update reports failure with error codes 0x80070002, 0x800F0922, 0x8007000D, etc.
- Update progress bar stays at 0% for an extended period.
- After restart, Windows shows "Undoing changes made to your computer" and rolls back the update.
- The Settings > Windows Update page displays "We couldn't complete the updates. Undoing changes."
- Laptop repeatedly restarts and gets stuck at the configuring updates screen.

Common causes

- Insufficient free space on the system drive (<10?GB).
- Corrupted Windows Update components (SoftwareDistribution, catroot2).
- Lack of network connectivity - VPN not connected or intermittent.
- Pending restart required from a previous update.
- Feature update deferral policies blocking the latest quality update.
- Disk errors or failing hard drive preventing file writes.

Preliminary Checks (User)

Before diving into troubleshooting, verify the basics that often cause update failures. Ensure the laptop is plugged in or has sufficient battery charge, and that you are connected to the Acme VPN (GlobalProtect) if you are off-campus. Check available storage on the C: drive - Windows Update needs at least 10-15?GB free space for temporary files. Finally, note any error code displayed in the Windows Update settings window; you will need it for the troubleshooter or when contacting the Service Desk.

Self-Help Procedure - Fix Windows Update Stuck/Failure

Follow these steps in order. If a step resolves the issue, you can stop and verify that updates install successfully. Each step is safe to run on a managed Acme laptop and does not require administrative privileges beyond what you already have.

1. 1. Run the Windows Update Troubleshooter: Open Settings -> System -> Troubleshoot -> Other troubleshooters -> Click "Run" next to "Windows Update". Follow the on-screen prompts and apply any fixes it suggests.
2. 2. Reset Windows Update components (requires admin): Open Command Prompt as Administrator and execute the following commands one by one, pressing Enter after each:

```
net stop wuauserv
net stop cryptSvc
net stop bits
net stop msiserver
ren C:\Windows\SoftwareDistribution SoftwareDistribution.old
ren C:\Windows\System32\catroot2 catroot2.old
net start wuauserv
net start cryptSvc
net start bits
net start msiserver
```

3. 3. Free up disk space: Open Settings -> System -> Storage -> Temporary files and remove items such as Windows

Update Cleanup, Temporary Internet Files, and Recycle Bin. Aim for at least 15?GB free on C:.
4. 4. Ensure VPN connectivity: Launch the GlobalProtect client, confirm you see "Connected" to vpn.acme.example. If you are on-site, make sure you are connected to the ACME-Corp Wi-Fi (802.1X).
5. 5. Retry the update: Return to Settings -> Windows Update -> Check for updates. Allow the process to run; it may take several minutes.
6. 6. If the update still fails, download the latest cumulative update manually: Go to the Microsoft Update Catalog (https://www.catalog.update.microsoft.com), search for your Windows 11 version (e.g., "Windows 11 version 23H2") and download the most recent "Cumulative Update" .msu file. Double-click the .msu to install, then restart when prompted.
7. 7. After restart, verify that the update succeeded by checking Settings -> Windows Update -> Update history. The status should show "Successful" for the latest KB article.

Service-Desk Procedure - Agent Workflow

When a user reports a Windows Update failure after attempting basic checks, follow this structured workflow. Use remote assistance tools (Teams screen share or Remote Help) to perform the steps, and document each action in the ticket.

1. 1. Verify user environment: Confirm device model, Windows 11 build, Intune compliance status, and that the laptop is encrypted (BitLocker) with the recovery key stored in Intune.
2. 2. Check free space on C: drive via File Explorer or the command `wmic logicaldisk where "DeviceID='C:'" get FreeSpace,Size`. If <10?GB, advise cleanup or temporarily move large files to OneDrive.
3. 3. Ensure VPN connection: Ask the user to launch GlobalProtect and verify the "Connected" state; if off-site and unable to connect, guide them through re-installing the client or checking their home ISP.
4. 4. Run the Windows Update Troubleshooter remotely (same as user step 1) and apply any recommended fixes.
5. 5. If the troubleshooter does not resolve the issue, reset update components using the admin command script provided in the self-help section (steps 2-4).
6. 6. Examine logs: Open `%windir%\Logs\CBS\CBS.log` and `%windir%\WindowsUpdate.log` (enable via `Get-WindowsUpdateLog` if needed). Look for error codes matching the user's symptom and note any recurring failures (e.g., 0x800F0922 indicating missing source files).
7. 7. Deploy the latest quality update via Intune: In the Endpoint Manager admin center, create or update a Windows 10/11 update ring to approve the most recent KB article, then force a sync (`Device > Sync`) and monitor installation status.
8. 8. If a feature update is being blocked by a deferral policy, temporarily adjust the policy for the device (or user group) to allow the feature update, install it, then revert the policy after successful completion.
9. 9. After the update installs, request a restart and verify update history shows success. Close the ticket with resolution notes including the final KB version installed and any actions taken (e.g., disk cleanup, component reset).

Verification and Confirmation

After applying either the self-help or service-desk steps, confirm that Windows Update is functioning correctly. Check that the system shows the latest build number (e.g., via `winver`), that the update history lists the installed KB with a successful status, and that no pending restarts remain. Advise the user to monitor for any unusual behavior over the next 24?hours and to report if the issue recurs.

When to escalate

If the update continues to fail after completing the service-desk procedure-particularly when logs indicate persistent component store corruption, disk errors, or suspected malware-escalate to the Desktop Support senior team or Application Support. Include in the escalation: the exact error code(s) from Windows Update, relevant excerpts from CBS.log and WindowsUpdate.log, screenshots of the update failure, a summary of troubleshooting steps already performed, and the device's serial number and Intune compliance state. For issues that appear to be related to network authentication or certificate problems (e.g., VPN connection failures despite correct configuration), involve the Network Support team after verifying GlobalProtect client logs.

Keywords: Windows Update, update stuck, 0% progress, error 0x80070002, error 0x800F0922, rolling back, troubleshooter, SoftwareDistribution, catroot2, disk cleanup, VPN, GlobalProtect, cumulative update, Microsoft Update Catalog, Intune, BitLocker, service desk, Desktop Support